

LLMs in practice: one case study

NL2SQL, RAG

SPEAKER

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ROLE

Staff ML engineer, Modulai

DATE

14 September 2026

What the next hour looks like

-
- 01 About me
 - 02 Modulai
 - 03 Master's thesis positions
 - 04 Case study
 - NL2SQL
 - RAG
-



About me

- From Riga, Latvia
- Moved to Sweden in 2021
- First batch of the UU Data Science MSc
- Banks → fintech → ML consulting
- Handpan, Wim Hof method



Wim Hof winter expedition, March 2025



■ Opinionated. No bullshit. Maximum impact.

JUNE · 2026

The AI Agency

who we are

We've been doing this since before it was obvious

An AI agency that advises, builds and runs tailored, intelligent systems, working from two studios in Stockholm and Gothenburg.

THE TEAM / 01



30+

ML engineers & researchers

modulai.io



150+

models in production



100+

clients, startup to enterprise



the gap

Most AI projects **stall** between research and production

Where most initiatives die is getting to a system that runs reliably in production and that your organisation can actually turn into results.



We are built to close that gap, moving from potential value to proven value at every step.



We are defined by our clients' success

OUTCOMES

6wks

A full investor relations AI platform in production

Swedish retail investor

300hrs

Of podcast audio transcribed, analysed and made searchable

Svenska Dagbladet

2M+

Customers served by a recommender in production

Lindex

SELECTED CLIENTS



open positions

We are open for master's theses

Check the website if you are interested.



June · 2026

Cases / portfolio.

Selected projects in applied AI and machine learning.

Finance

Retail

Media

Life sciences

Logistics

Gaming



CLIENT
Novo Nordisk

Trusted answers from regulated trial data

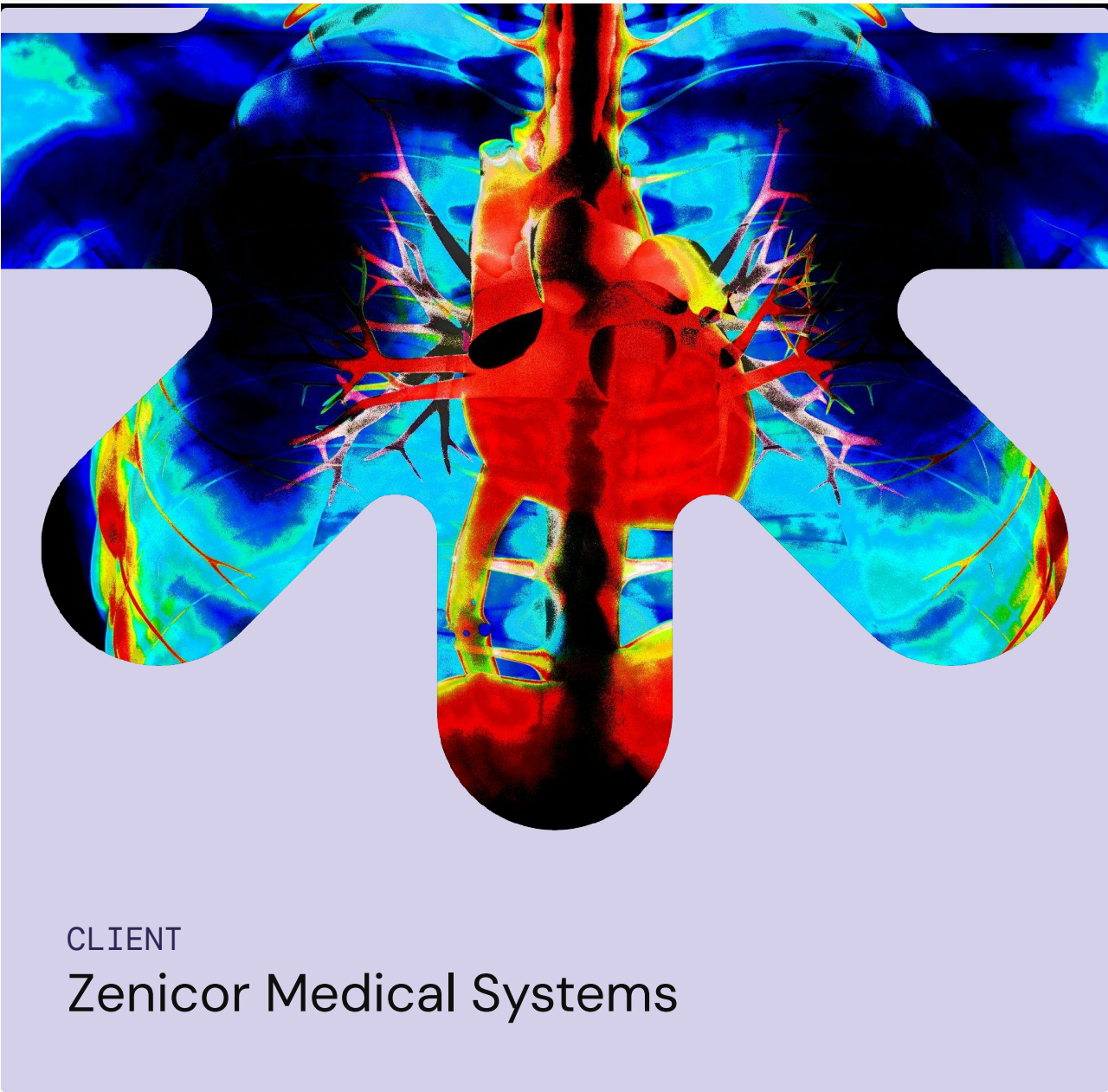
THE CHALLENGE	Information about clinical trials sits scattered across databases, reports, and notes. Finding a design detail and the reasoning behind it means cross checking several systems by hand, and in a regulated setting every answer has to be traceable and compliant.
THE SOLUTION	Modulai built a system of cooperating agents that plans each question, gathers the relevant sources, runs the analysis, and answers with citations back to the evidence. It can reply in plain text, build a table, or export aggregated data, with guardrails that keep it in scope and aligned with regulatory standards.
VALUE CREATED	Teams get faster, dependable access to clinical trial insight without manual cross checking, and every answer can be traced back to its source, which is essential in a regulated environment.

BUSINESS VALUE

- Process efficiency
- Risk reduction

CAPABILITIES

- GenAI / agents
- NLP
- RAG



CLIENT
Zenicor Medical Systems

Detecting hidden atrial fibrillation before a stroke

THE CHALLENGE	Paroxysmal atrial fibrillation comes and goes, so a single heart recording can look completely normal even when the condition is there. Thousands of people have a stroke every day from a case that was never caught.
THE SOLUTION	Working alongside heart specialists at Karolinska Institutet, Modulai learned from a large store of unlabelled heart recordings, shaped a new method for the problem, and tuned it to spot the hidden condition. The team also traced how the model reached each result, which matters when it informs a clinical decision.
VALUE CREATED	Learning from the unlabelled recordings lifted detection by 7 percent over the earlier approach, the step from a model of little practical use to one that catches the condition in recordings that look normal. That points to earlier diagnosis and fewer missed cases.

METRICS

+7%

better detection than before

BUSINESS VALUE

New business

Risk reduction

CAPABILITIES

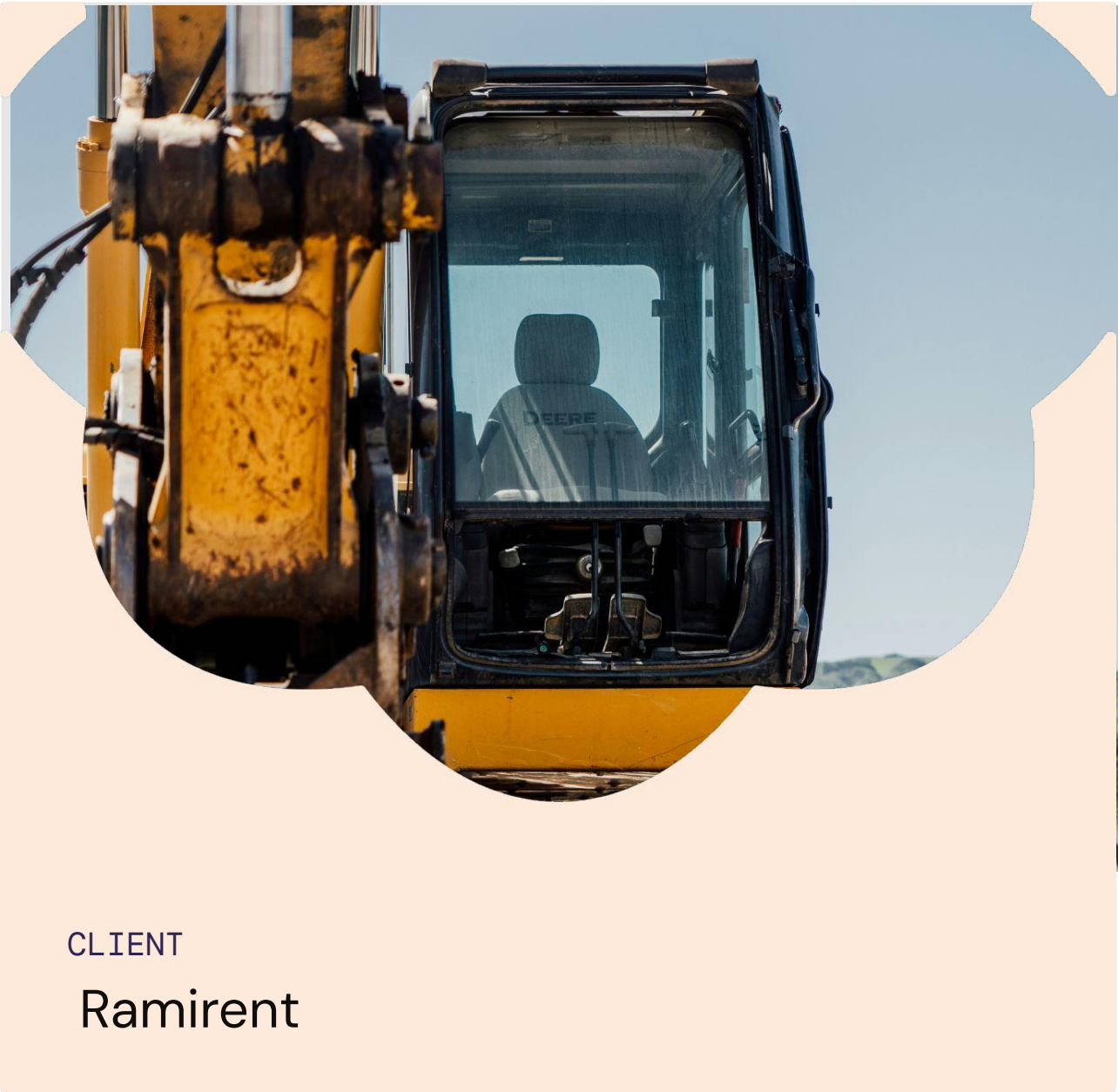
Deep learning

Time-series classification



Predicting when rented equipment will get returned

THE CHALLENGE	A rental company never quite knows when lent out equipment will return, which makes it hard to plan where machines should be and leads to needless transport between shops and warehouses.
THE SOLUTION	Modulai built a model that forecasts when each piece of equipment is likely to come back, giving the company a clearer view for planning.
VALUE CREATED	Better timing means smarter logistics and less unnecessary transport between local shops and the main warehouse.



CLIENT
Ramirent

BUSINESS VALUE

- Cost saving
- Process efficiency

CAPABILITIES

- Tabular ML
- Time-series classification



Predicting pharma shipment failures before they happen

THE CHALLENGE

The client is one of the world's leading providers of tracking for sensitive pharmaceutical goods in transit. Every shipment is followed minute by minute by a temperature logger, and if the goods spend too long above safe temperatures by the time they arrive, the whole shipment is rejected.

THE SOLUTION

Modulai built models from the temperature readings and supporting data of hundreds of thousands of past shipments. They estimate the risk that a shipment will fail, which guides the choice of packaging and route, and they form the basis of a warning system that alerts operators to rising risk along the way. The work was delivered as a web app for internal use.

VALUE CREATED

The client can now see the risk of a failed shipment before it leaves, choosing packaging and routes that keep goods safe, and operators can be warned while a shipment is still moving rather than finding out only when it is rejected on arrival.



CLIENT
TSS group

METRICS

20K+

100K+

30+

users worldwide

historical missions

years of experience

BUSINESS VALUE

Cost saving

Risk reduction

CAPABILITIES

Tabular ML

Time-series

Bayesian modeling

Deep learning



CLIENT
Fishbrain

Forecasting when and where the fish will bite

THE CHALLENGE	Fishbrain's millions of anglers want to know when and where their next trip is worth taking. The answer depends on weather, water, season, and countless local conditions.
THE SOLUTION	Modulai built a model from 2.5 million recorded catches, each tagged with time, place, species, and detailed weather, then set against a global climate model. It predicts the chance of landing each species through the day and powers BiteTime, the app's fishing forecast.
VALUE CREATED	BiteTime went live inside the app, putting a catch forecast in the hands of more than eleven million anglers before they ever head out.

METRICS

11M+

app users

2.5M

catches trained on

BUSINESS VALUE

New business

Revenue increase

CAPABILITIES

Tabular ML

Feature engineering

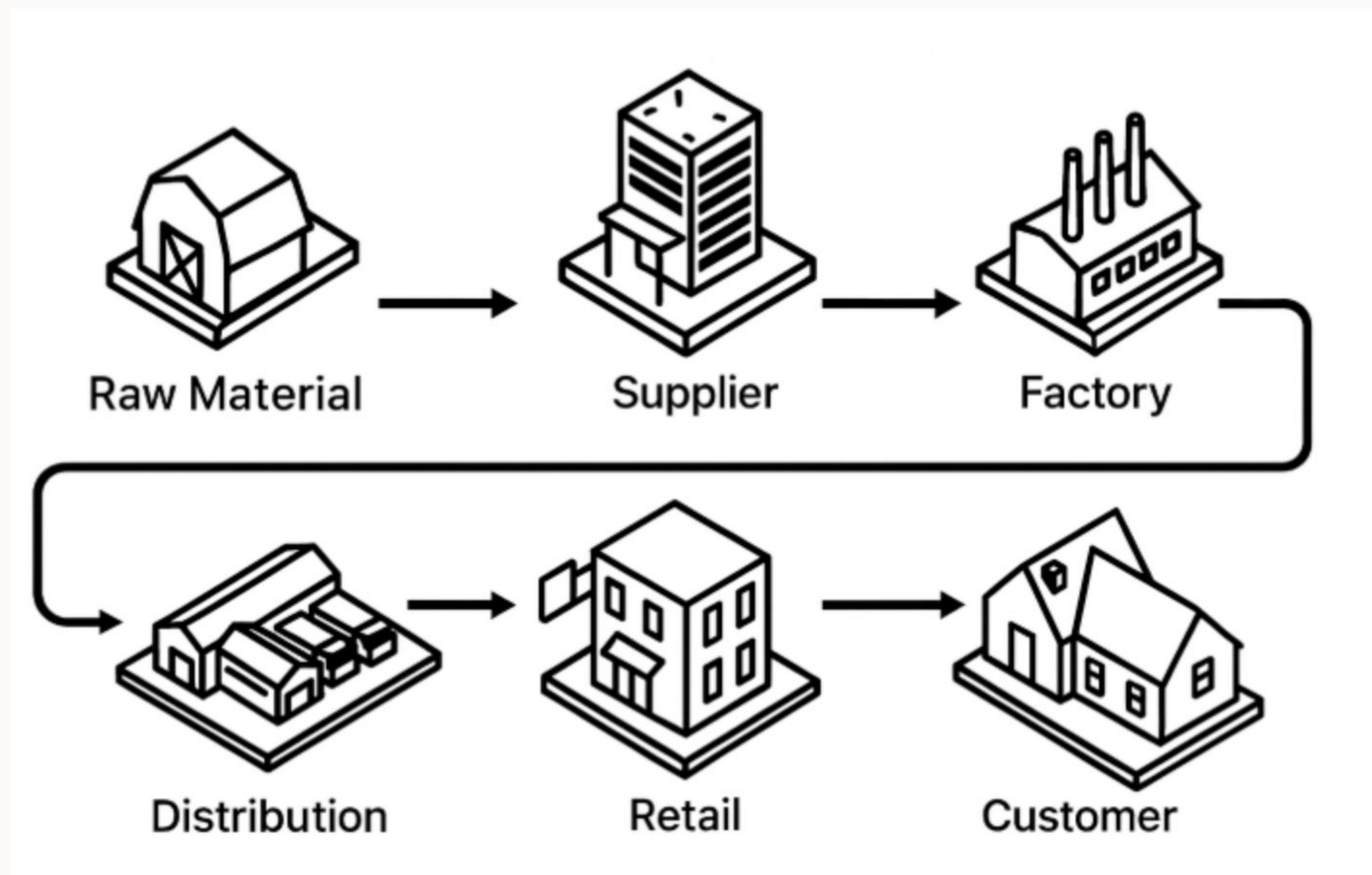
Deep learning

One case, two halves

NL2SQL and RAG are two branches of the same assistant inside one product.



Supply chain management



How do you make it efficient?

TOOLS LIKE

Scenario analysis Optimization

... WHICH REQUIRE

Large volumes of data A database-like structure



Optilogic

optilogic.com

A platform where supply chain teams model their network and ask what-if questions of it.

SCENARIO ANALYSIS

“If this factory gets closed, how does it hit production?”

OPTIMIZATION

“What is the cheapest way to produce, store and distribute across Sweden?”

All of it sits in a relational database — hundreds of tables describing the network.



The data speaks SQL. The users don't.

THE DATA

A large relational model

Facilities, products, demand, costs, lanes, constraints — all joined together.

≠

THE USERS

Supply chain professionals

They know the network better than anyone. They are not database experts, and they should not have to be.

So: let them ask in their own words, and let the system write the SQL.



in the product

Ask in words, get SQL back

The user describes the scenario they want. The system writes the query, shows it, and waits — the SQL runs only when the user clicks.

WHY IT MATTERS

A visible query is a reviewable query.

D

Create a scenario where a storage capacity in Atlanta in just 20%



SQL Query

```
UPDATE Facilities SET StorageCapacity = 0.2 * StorageCapacity::DOUBLE PRECISION WHERE City = 'Atlanta'
```

Run SQL

Scenarios

Scenario Name	reduce_storage_capacity_atlanta
Item Name	reduce_storage_capacity_atlanta
Table Name	Facilities
Action	storagecapacity = 0.2 * storagecapacity
Condition	city = 'atlanta'

Create Scenario





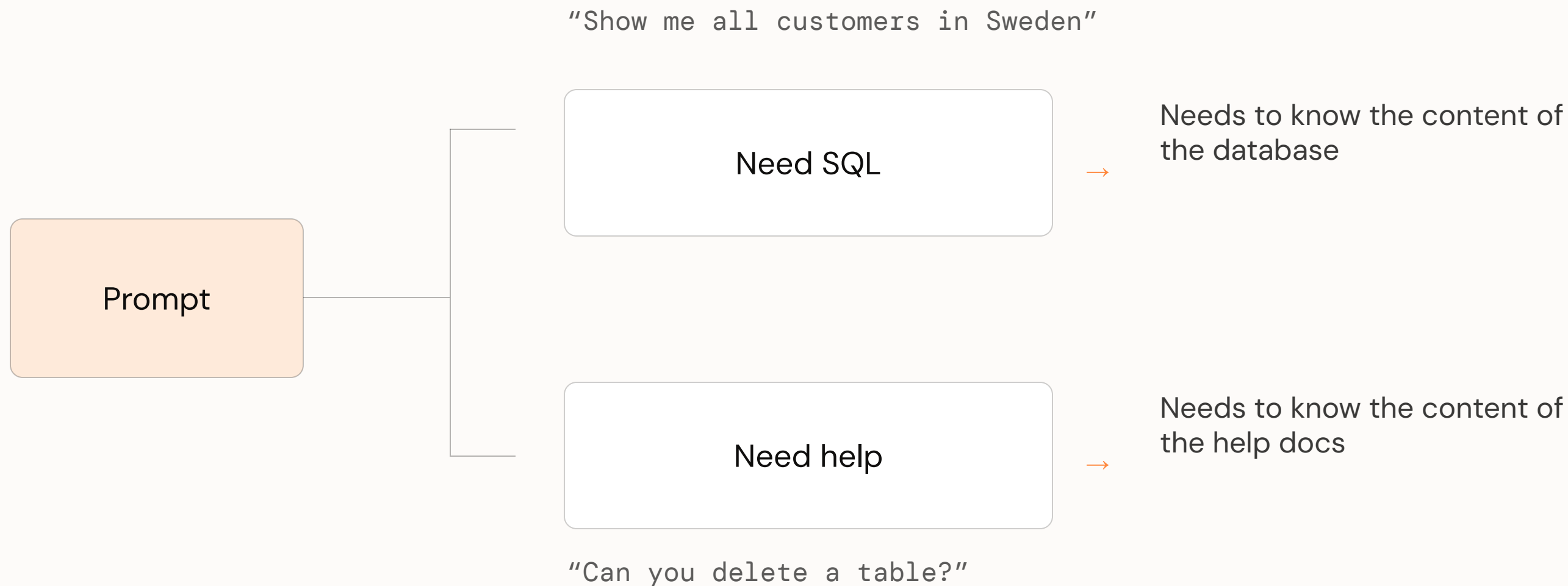


NL2SQL

Natural language in, SQL out.



One chat box, two kinds of question



REQUIRES

Function / tool calling

You describe the available functions; the model returns which one to call and with what arguments. That return value is the routing decision.

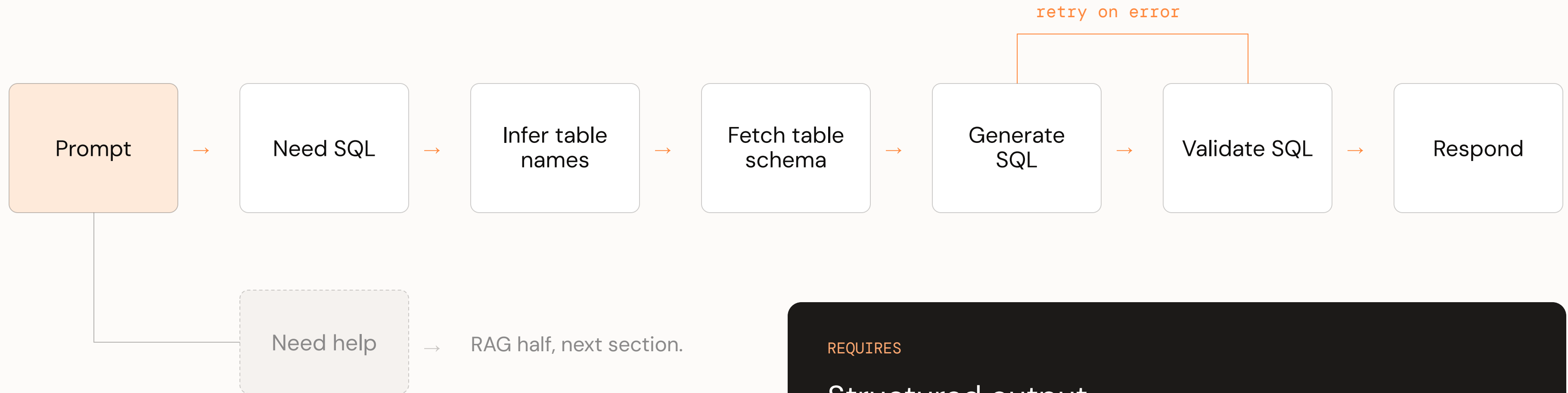
THE SPEC, LIVE

[platform.claude.com → tool use](https://platform.claude.com/tool-use)

DEMO Routing



The SQL path, end to end



REQUIRES

Structured output

SQL, table list and explanation come back as fields, not prose.

Model

The right model, plus few-shot examples and/or fine-tuning for complex SQL and house style.

DEMO Structured output, few-shot learning

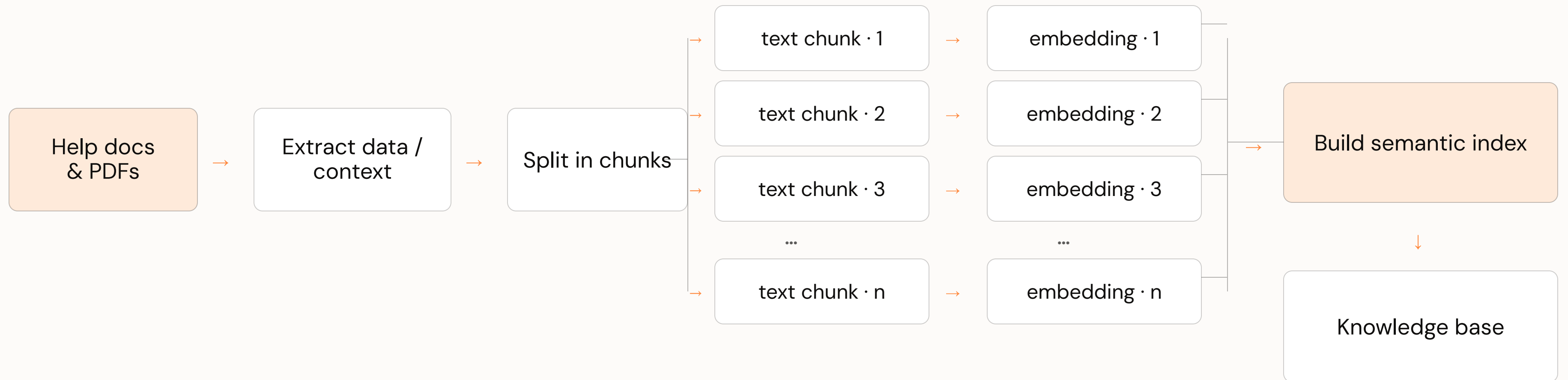
RAG

Retrieval-augmented generation — where the LLM is critical, but not the bottleneck



Ingestion

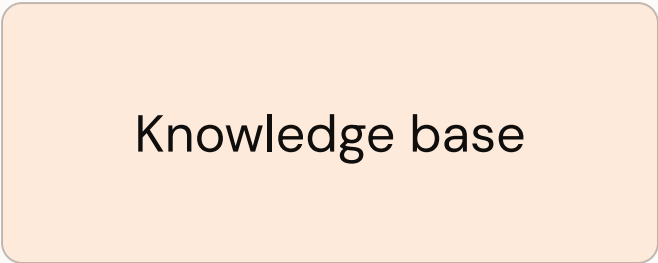
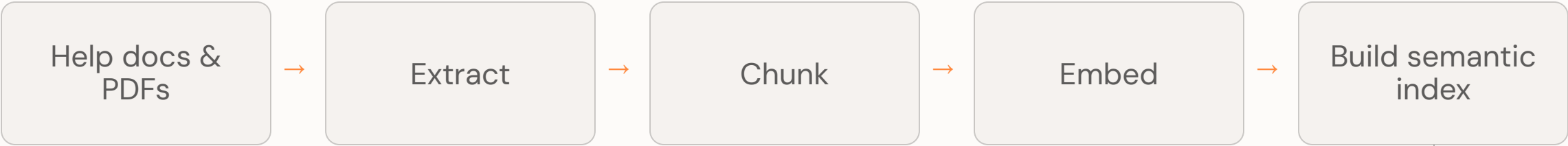
Runs offline. Sets the ceiling on what can ever be answered.



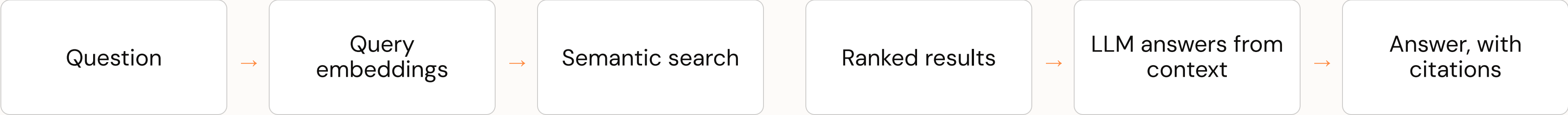
Querying

DEMO End-to-end RAG

INGESTION · OFFLINE



QUERY · ONLINE

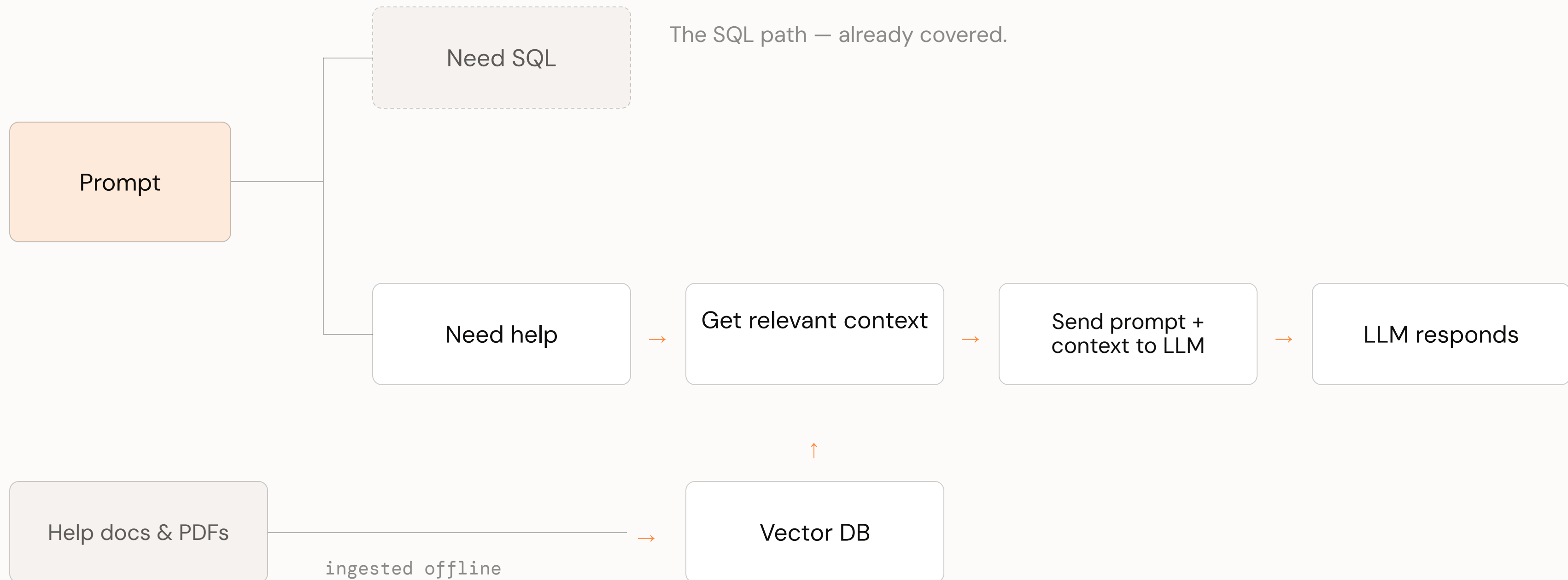


Closing the circle

The help branch is RAG, running on the index we just built.

THOUGHTS

Retrieval is a bottleneck for generation
Domain adaptation



questions

Thank you!

Questions?

modulai.io

